

Total No. of Questions : 4]

SEAT No. :

PF230

[Total No. of Pages : 2

APR-26/SE/Insem-284

S.E. (Artificial Intelligence and Machine Learning) (Insem)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - IV) (218555)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.No.1 or Q.No.2 and Q.No.3 or Q.No.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Consider a line P (2, 6) to Q (10, 12). Use Bresenham's line drawing algorithm to rasterize the line from P to Q. Draw the pixel wise rasterization of Line. **[8]**

- b) What is computer graphics? Define: **[7]**
- i) Frame Buffer
 - ii) Aspect Ratio
 - iii) Resolution
 - iv) Persistence

OR

Q2) a) Consider a line from P (5, 5) to Q (13, 10). Use DDA line drawing algorithm to rasterize the line from P to Q. Draw the pixel wise rasterization of Line. **[8]**

- b) Explain display file structure. Why is display file interpreter used? Which are the commands used in display file interpreter. **[7]**

Q3) a) Apply following transformations on polygon A (10, 10), B (20, 40), C (50, 40), D (40, 0) **[8]**

- i) Translation 10, 20 units along X & Y directions.
 - ii) Rotate 45 degrees about the origin.
Reflection against X-axis.
- b) Explain concave and convex polygons with diagram. Explain even-odd method for testing a pixel inside or outside the polygon. **[7]**

OR

P.T.O.

- Q4)** a) What are the steps involved in scan line polygon filling algorithm. [8]
b) Apply the shearing transformation to square with given coordinates below. [7]
A(0,0), B(0,3), C(3,3), D(3,0)
i) Shear Parameter value of 0.7 relative to the line $Y_{ref} = -1$.
ii) Shear Parameter value of 0.7 relative to the line $X_{ref} = -1$.

